SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING **COURSE CODE: CSA43**

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. **(L2)**

Assessment Tool: Code along exercise: Make your first HTML page together with CSS

Weightage: 10%

QUESTIONS: **Total Marks: 10**

Code of Conduct:

I _R.Jashwanth(192411042)____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

R.Jashwanth

Scenario: Web Development Standards Implementation

A start up is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols.
The following observations are made:

- The team needs to embed a promotional video on the homepage.
- A registration form must securely send user data to the server.
- The navigation menu must follow semantic HTML standards.
- XML configuration files are being used but contain syntax errors.
- The application must communicate over the correct web protocol.

Data Snippets Provided

Snippet 1: Video Embedding Options

A. <media src="video.mp4"></media>
B. <video src="video.mp4" controls></video>
C. <movie src="video.mp4"></movie>
D. <vid src="video.mp4"></vid>

Snippet 2: Form Submission Methods

A. GET
B. POST
C. PUT
D. DELETE

Snippet 3: Navigation Elements

A.
B. <nav>
C. <section>
D. <header>

Snippet 4: XML Syntax

A. <note><to>User</to><from>Admin</from>

B. <note><to>User</to><from>Admin</from></note>

C. <note><to>User</from></note>

D. <note to="User"></note

Snippet 5: Web Communication Protocols

A. FTP

B. HTTP C. SMTP

D. SNMP

Questions:

1. Select the **correct HTML tag for video embedding** and justify your answer.
2. Identify the **appropriate HTTP method for form submission** and explain why.
3. Choose the **proper semantic element for navigation** and justify its usage.
4. Identify the **correct XML syntax** and explain the error in other options.
5. Choose the **correct protocol for web communication** and justify your choice.

Introduction

Web development standards help developers create websites that are structured, accessible, secure, and reliable. HTML is used to structure web pages, HTTP methods handle communication between clients and servers, XML stores and organizes data, and web protocols enable communication over the internet. This assignment explains the correct standards to be followed in a modern web application.

Q1. Select the correct HTML tag for video embedding and justify your answer.

Answer: Option B — <video src="video.mp4" controls></video>

Explanation:

The <video> tag is the standard HTML element used to embed video content into a web page. The src attribute specifies the video file, while the controls attribute displays playback options such as play, pause, and volume.

Example:

<video src="video.mp4" controls></video>

The other options, <media>, <movie>, and <vid>, are not standard HTML elements for embedding videos. Therefore, Option B is correct.

Q2. Identify the appropriate HTTP method for form submission and explain why.

Answer: Option B — POST

Explanation:

The POST method is appropriate for submitting registration form data to a server. It sends the submitted data in the HTTP request body rather than displaying it in the URL, as GET commonly does.

Example:

```
<form action="/register" method="POST">
  <input type="text" name="username" required>
  <input type="email" name="email" required>
  <input type="password" name="password" required>
  <button type="submit">Register</button>
</form>
```

POST is suitable for registration because it sends the submitted information in the request body. However, POST alone does not provide encryption. **HTTPS must be used to protect sensitive information during transmission.**

Q3. Choose the proper semantic element for navigation and justify its usage.

Answer: Option B — <nav>

Explanation:

The <nav> element is a semantic HTML element used to define a section containing major navigation links. These links help users move between different pages or sections of a website.

Example:

```
<nav>
  <a href="index.html">Home</a>
  <a href="about.html">About Us</a>
  <a href="services.html">Services</a>
  <a href="contact.html">Contact</a>
</nav>
```

The <nav> element improves the readability and organization of HTML code and helps assistive technologies identify navigation sections. Therefore, Option B is correct.

Q4. Identify the correct XML syntax and explain the error in other options.

Answer: Option B

Correct XML code:

```
<note>
  <to>User</to>
  <from>Admin</from>
</note>
```

Explanation:

Option B follows the basic XML syntax rules. Every opening tag has a corresponding closing tag, and the elements are properly nested within the root element <note>.

Errors in the other options:

- **Option A:** The <note> element is not closed, making the XML incomplete.
- **Option C:** The <to> element is incorrectly closed with </from>, resulting in mismatched tags.
- **Option D:** The opening <note> tag is incomplete because the closing angle bracket (>) is missing.

Therefore, Option B is the correct XML syntax.

Q5. Choose the correct protocol for web communication and justify your choice.

Answer: Option B — HTTP

Explanation:

HTTP stands for Hypertext Transfer Protocol. It is an application-layer protocol used for communication between web browsers and web servers. It allows clients to request web pages and other resources, while servers return the requested resources and responses.

Example: When a user opens a website, the browser sends an HTTP request to the web server, and the server responds with the requested web content.

The other protocols have different purposes:

- **FTP (File Transfer Protocol):** Used to transfer files between systems.
- **SMTP (Simple Mail Transfer Protocol):** Used to send email messages.

- **SNMP (Simple Network Management Protocol):** Used to monitor and manage network devices.

For secure communication, HTTPS should be used because it protects data transmitted between the browser and server through encryption.

Therefore, Option B is correct.

Conclusion

Following web development standards is essential for building a reliable and secure web application. The HTML <video> tag supports video embedding, the POST method is suitable for registration form submission, and the <nav> element provides semantic navigation. Correct XML syntax ensures well-formed configuration files, while HTTP enables communication between web browsers and servers. Using HTTPS further improves the security of web communication.

Formatting for submission: Use Times New Roman, 12-point font for the main text, 14-point bold for question headings, 1.5 line spacing, justified alignment, and 1-inch margins. Keep the main title centered and bold.

RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, wellstructured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand